Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Invert sugar is a mixture of

(A) glucose and sucrose

(B) glucose and fructose

(C) glucose and maltose

(D) lactose and maltose

Q.2

Which of the following is a linear polymer?

(A) Amylopectin

(B) Glycogen

(C) Starch

(D) Amylose

Q.3

Monomer of cellulose is

(A) lactose

(B) maltose

(C) fructose

(D) glucose

Q.4

The number of amino acids in insulin is

(A) 31

(B) 41

(C) 51

(D) 61

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Q.5

The protein molecule present in haemoglobin is

(A) thrombin

(B) fibrinogen

(C) globulin

(D) casein

Q.6

Which of the following bases is not common to DNA and RNA?

(A) Adenine

(B) Guanine

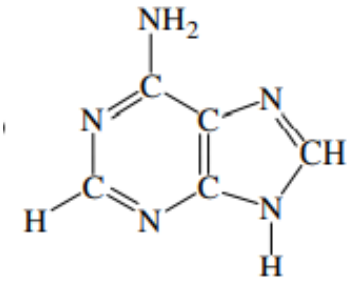
(C) Thymine

(D) Cytosine

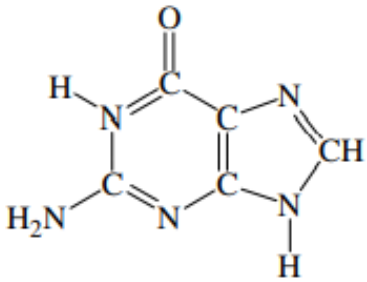
Q.7

Which of the following structures represents guanine?

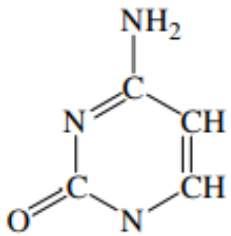
(A)



(B)

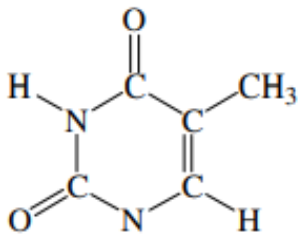


(C)



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(D)



Q.8

Which of the following is the unsaturated fatty acids?

(A) Palmitic acid

(B) Stearic acid

(C) Oleic acid

(D) Lauric acid

Q.9

The artificial sweetener that has the highest sweetness value in comparison to cane sugar is:

(A) Sucralose

(B) Aspartame

(C) Saccharin

(D) Alitame

Q.10

Which of the following enzymes converts glucose into ethanol?

(A) Maltase

(B) Diastase

(C) Zymase

(D) Invertase

Q.11

The disease pernious anaemia is caused by the deficiency of vitamin

(A) A

(B) K

(C) B_1

(D) B_{12}

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Q.12

Which of the following statements is correct?

(A) A polymer of α-Glucose is readily digested by human beings and not that of β-glucose

(B) A polymer of β-Glucose is readily digested by human being and not that of α-glucose

(C) Polymers of both α - and β-glucoses are readily digested by human beings

(D) Polymers of both α - and β-glucose are not readily digested by human beings

Q.13

Which of the following is not correct?

(A) Vitamins are synthesized within the body

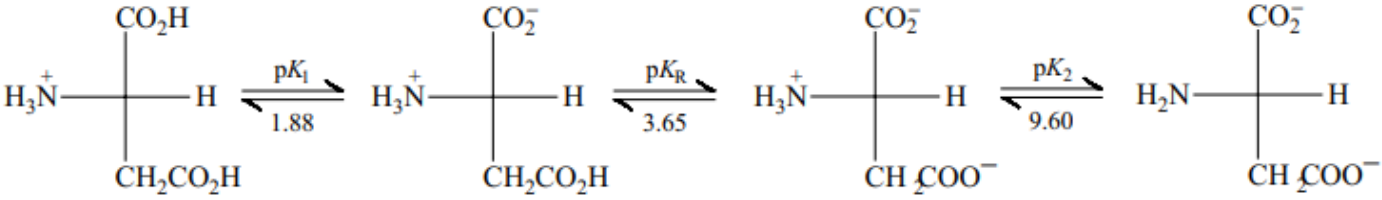
(B) Hormones are synthesized within the body

(C) Hormones are also known as chemical messengers

(D) Hormones are hightly potent and so are produced in small quantities

Q.14

Consider the following sequence for aspartic acid:



The pI (isoelectric point) of aspartic acid is:

(A) 3.65

(B) 2.77

(C) 5.74

(D) 1.88

Q.15

Observation of "Rhumann's purple" is a confirmatory test for the presence of

(A) Starch

(B) Reducing sugar

(C) protein

(D) Cupric Ion

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Q.1 Answer:

glucose and fructose

Solution:

Invert sugar is a mixture of glucose and fructose.

Q.2 Answer:

Amylose

Solution:

Amylose is a linear polymer.

Q.3 Answer:

glucose

Solution:

Glucose is a monomer of cellulose.

Q.4 Answer:

51

Solution:

There are 51 amino acids in insulin.

Q.5 Answer:

globulin

Solution:

globin molecules

Hemoglobin consist of protein subunits (the globin molecules), and these proteins, in turn, are folded chains of a large number of diferent amino acids called polypeptides

Q.6 Answer:

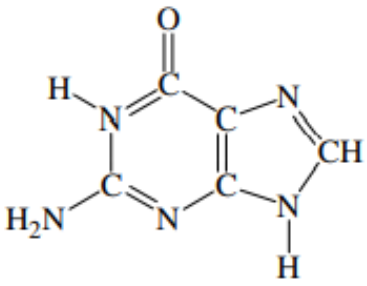
Thymine

Solution:

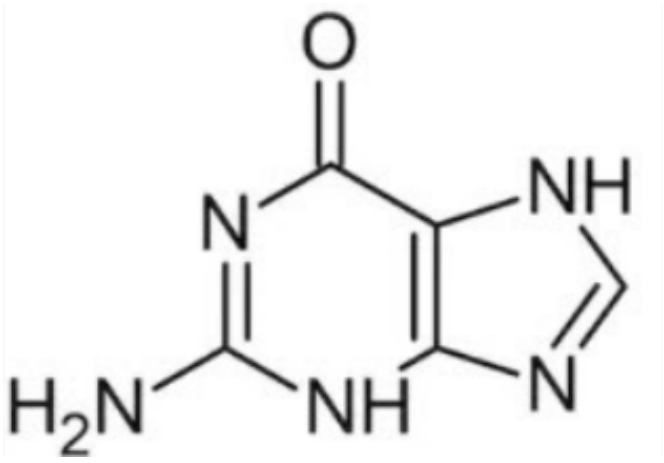
Pyrimidines include Thymine, Cytosine, and Uracil bases which are denoted by letters T, C, and U respectively. Thymine is present in DNA but absent in RNA, while Uracil is present in RNA but absent in DNA. Cytosine is present in both DNA and RNA.

Q.7 Answer:

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Solution:



Q.8 Answer:

Oleic acid

Solution:

Saturated monocarboxylic fatty acids have general formula $C_nH_{2n}O_2$
Palmitic acid is a saturated fatty acid with the formula $C_{16}H_{32}O_2$
Lauric acid is a saturated fatty acid with the formula $C_{12}H_{24}O_2$
Stearic acid is a saturated fatty acid with the formula $C_{18}H_{36}O_2$.
Linoleic acid is an unsaturated fatty acid with formula $C_{18}H_{32}O_2$

Q.9 Answer:

Alitame

Solution:

The relative sweetness with respect to sucrose are as follows
Sucralose (600)
Aspartame (100)
Saccharin (550)
Alitame (2000)

Q.10 Answer:

Zymase

Solution:

Fermentation process is catalyzed by enzyme Zymase which converts sugar to ethanol and carbon dioxide.

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So the correct option is 'Zymase.'

Q.11 Answer:

B_{12}

Solution:

Either a lack of vitamin B-12 or a lack of folate causes a type of anemia called megaloblastic anemia (pernicious anemia). With these types of anemia, the red blood cells don't develop normally.

Q.12 Answer:

A polymer of α-Glucose is readily digested by human beings and not that of β-glucose

Solution:

α-Glucose (starch) is readily digested by human beings but a polymer of β-glucose (cellulose) is not digestable.

Q.13 Answer:

Vitamins are synthesized within the body

Solution:

Vitamins are not synthesized within the body.

Q.14 Answer:

2.77

Solution:

At the isoelectric point of aspartic acid, we have

$$\left[\mathrm{H_3NCH\left(CH_2COOH\right)CO_2H}\right]=\left[\mathrm{H_3\overset{+}{N}CH\left(CH_2COO^-\right)CO_2^-}\right]$$

$$\mathrm{pI}=\frac{1}{2}\left(\mathrm{p}K_1+\mathrm{p}K_\mathrm{R}\right)=\frac{1}{2}(1.88+3.65)=2.77$$

Q.15 Answer:

protein

Solution:

Rhumann's purple is obtained when ninhydrin is used to detect ammonia or primary and secondary amines. The amine group in amino acid of proteins shows Rhumann's purple test.

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